Draft version June 28, 2026
Typeset using LaTeX twocolumn style in AASTeX631

Time-Resolved Empirical Spectroscopy of a Large Sample of Single-Pulse Gamma-Ray Bursts: Thermal versus Two-Break Origins of the Spectral Curvature

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ABSTRACT

Gamma-ray bursts (GRBs) are highly variable and exhibit strong spectral evolution, and the physical origin of their prompt emission remains debated. Here we present a time-resolved spectral analysis of a uniform sample of 106 single-pulse, bright Fermi/GBM bursts, comprising 1057 time-resolved spectra, fit with a family of empirical photon models—the Band function, a cutoff power law, a smoothly broken power law, a double smoothly broken power law (2SBPL), and the two non-thermal continua each with an added blackbody. We use empirical models as a model-agnostic probe and ask whether the thermal–non-thermal $E_{\rm p}$–kT correlation reported by Burgess et al. (2014a) for six bursts appears in a large sample, and what additional spectral properties emerge. We find that the low-energy photon index peaks at $\alpha \approx -0.7$, between the slow- and fast-cooled synchrotron limits ($-2/3$ and $-3/2$); that 91% (151/166) of the time bins requiring curvature beyond a single break are equally or better described by adding a blackbody than by an explicit second break (the 9% two-break fraction is a lower limit); and that where a blackbody is decisively required (ΔAIC ≥ 10), the Burgess et al. (2014a) correlation is recovered in 89% (17/19) of testable bursts, most strikingly in GRB 130427A ($\rho = 0.93$, $E_{\rm p} \propto kT^{0.9}$). We further find a positive correlation between the two synchrotron break energies ($\rho = 0.57$; intrinsic scatter 0.38 dex), present both within and between bursts. We caution, however, that no single empirical model is decisively preferred over the next-best alternative in 92% (875/950) of comparable bins, and that an empirical continuum can absorb a sub-dominant thermal component. We discuss the implications for the prompt radiation mechanism and jet composition.

Keywords: Gamma-ray bursts (629) — Spectroscopy (1558) — Non-thermal radiation sources (1119) — Astrostatistics (1882)

1. INTRODUCTION

Gamma-ray bursts (GRBs) are highly variable and exhibit strong spectral evolution: during the prompt phase—the initial seconds to minutes of keV–MeV emission—the νF_ν peak energy $E_{\rm p}$ (the energy at which the burst radiates most of its power) and the low-energy photon index α (the power-law slope of the photon spectrum below the peak) typically vary with the prompt light curve (e.g., Norris et al. 1996; Kaneko et al. 2006; Lu et al. 2012; Yu et al. 2016). To first order the prompt spectrum is well described by the empirical Band function (Band et al. 1993) or a cutoff power law, yet these functions are physically agnostic: the same curvature can arise from distinct emission mechanisms, and the fitted parameters do not by themselves identify the radiative process (e.g., Kumar & Zhang 2015).

Two physically motivated pictures dominate. In the first, part of the prompt flux is quasi-thermal emission from the jet photosphere—the radius at which the relativistic outflow becomes transparent to its own photons, much as the solar photosphere defines the visible surface of the Sun—approximated by a blackbody (BB) superposed on a non-thermal continuum (e.g., Mészáros & Rees 2000; Ryde 2004; Pe'er et al. 2007; Guiriec et al. 2010, 2011). Fitting a physical synchrotron-plus-blackbody model to six bright single-pulse bursts, Burgess et al. (2014a) reported a tight per-burst correlation $E_{\rm p} \propto kT^{\alpha}$; because both energies are set by conditions near the base of the outflow, the index of this correlation traces how the jet accelerates, with $\alpha \sim 1$ indicating a baryon-dominated (fireball) and $\alpha \sim 2$ a magnetically-dominated jet. In the second picture, the

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curvature is optically-thin synchrotron radiation from a cooling electron population: electrons accelerated to a characteristic energy radiate at the injection frequency ν_m, and as they lose energy their emission extends down to the cooling frequency ν_c—the energy at which an electron radiates away its energy within the duration of the emission episode—so the spectrum breaks at *two* characteristic energies (Sari et al. 1998). A double smoothly broken power law (2SBPL) captures both breaks and has been used to argue for a synchrotron origin (e.g., Oganesyan et al. 2017, 2018; Ravasio et al. 2018, 2019). Most recently, Mei et al. (2025) found that, under the synchrotron interpretation, the fundamental spectral–energetics relation is between ν_c and the isotropic luminosity rather than the classical $E_{\rm p}$–$L_{\rm iso}$ relation (the Yonetoku relation; Yonetoku et al. 2004).

Crucially, the two pictures produce similar curvature near the spectral peak: over the finite GBM band, the bump added by a blackbody to a single-break continuum bends the total spectrum in nearly the same way as a second power-law break, so a BB+SBPL composite is an effective *proxy* for a 2SBPL. Distinguishing them, and characterizing the resulting spectral properties at scale, is the goal of this work. Single-pulse bursts offer the cleanest test, since a single, non-overlapping pulse exhibits the simplest spectral evolution and avoids the blending of distinct emission episodes (e.g., Ryde & Pe'er 2009; Basak & Rao 2014; Burgess et al. 2014b; Busby & Lazzati 2024). Basak & Rao (2014) performed exactly such a time-resolved study of nine single-pulse Fermi bursts with thermal-family models, finding that a blackbody peaks well below the Band νF_ν peak and that α frequently exceeds the synchrotron limit, especially in hard-to-soft pulses; Burgess et al. (2014a) were limited to six bursts. Whether these results are general has not been tested on a large, uniformly-analyzed sample.

In this paper we take a deliberately empirical approach. We fit a family of model-agnostic functions uniformly to a large, single-pulse sample, and we investigate (i) the distributions and time evolution of the spectral parameters, (ii) the correlations among them, (iii) whether the Burgess et al. (2014a) $E_{\rm p}$–kT correlation appears in a large sample, and (iv) what additional spectral properties—in particular a relation between the two synchrotron breaks—emerge. The paper is organized as follows. Section 2 describes the sample, data reduction, and spectral fitting; Section 3 presents the observed parameter distributions, spectral evolution, and correlations; Section 4 assesses the compatibility with emission models; Section 5 discusses the results; and Section 6 summarizes our findings. Uncertainties are quoted at the 1σ (68%) level unless stated otherwise.

2. METHODOLOGY

2.1. *Initial Burst Selection*

We select single-pulse bursts from the Fermi/GBM catalog (Meegan et al. 2009; Gruber et al. 2014) using the shape-based "horizontal-line" algorithm of Busby & Lazzati (2024), which scores how well a single FRED-like (fast-rise, exponential-decay) pulse describes the background-subtracted light curve. We retain bursts above the Busby & Lazzati single-pulse threshold (score $\gtrsim 0.9983$) and apply a fluence cut of $>10^{-5}$ erg cm^{-2} (10–1000 keV, GBM catalog) to ensure adequate time-resolved signal-to-noise, since a robust spectral shape requires sufficient counts per bin (Section 2.3). No T_{90} restriction is imposed; the sample therefore contains both short- and long-duration bursts whose emission is dominated by a single pulse. We emphasize that this selection is on pulse *shape*, not brightness: the sample fluence spans a factor of $\sim$250. The final sample comprises 106 bursts (Table 2).

2.2. *Detector, Source, and Background Selections*

For each burst the detector–source angle θ of the 14 GBM detectors is computed from the spacecraft attitude (the POSHIST quaternions interpolated to the trigger time). We use the time-tagged event (TTE) data of the NaI detectors with $\theta \leq 50°$; a detector at $50° < \theta \leq 60°$ is also kept if it appears in the trigger's on-board (BCAT) detector mask—it triggered, so the source lay in its field of view—and if no NaI then qualifies the single closest BCAT NaI is retained. Up to the three lowest-angle NaI are used, each paired with the BGO on its side of the spacecraft (b0 for n0–n5, b1 for n6–nb). For the 11 bursts with usable LAT Low-Energy (LLE) data (a subset of the 18 LAT-detected bursts) we add the LLE data above 30 MeV (Atwood et al. 2009). Each detector's response matrix is used; for multi-epoch `rsp2` responses the matrix covering the source interval is taken. We use 8–900 keV for NaI, masking the 33–40 keV region around the iodine K-edge—a sharp jump in the crystal's absorption efficiency at 33.17 keV that is difficult to calibrate (Ronchi et al. 2020)—with 0.3–40 MeV for BGO and 30–100 MeV for LLE. Inter-detector calibration is handled by free effective-area constants in the range 0.8–1.2, frozen to unity for the reference NaI (as in Ronchi et al. 2020). The source interval is set to bracket the burst emission (Section 2.3). Each selected detector receives its own pre- and post-burst background intervals ($\sim$50–150 s each), fit with a time polynomial of order $\leq$3—the order chosen by a likelihood-ratio test—and interpolated beneath the burst: the detector background varies smoothly with the spacecraft's orbital position, so windows on either side of the burst constrain its level during the burst itself. An accurate background is the prerequisite for recovering any sub-dominant spectral component, so every window is reviewed and approved—by a human at an interactive light-curve display, or by a vision-capable agent—with each approval recorded together with its origin, so the provenance of every selection is auditable.[1]

2.3. *Light-curve Binning*

Within the approved emission interval of each burst, we define the time bins from the reference NaI (the lowest-angle approved NaI) using the Bayesian Blocks algorithm (Scargle et al. 2013) applied to its background-subtracted 8–900 keV count rate (the `3ML` implementation, with the polynomial background included and a false-alarm prior $p_0 = 0.01$). Bayesian Blocks partitions a light curve at the points where the count rate changes significantly, producing bins that are long during quiet stretches and short across sharp variations, so that each bin samples an approximately constant spectral state. We then impose a per-block significance floor: each block's net significance is evaluated with the modeled-background (Gaussian) Li & Ma statistic, and blocks below 5σ are dropped at the leading and trailing edges and merged into a neighbour in the interior until every surviving block clears 5σ, so that each time bin is simultaneously a distinct count-rate state and a statistically adequate spectrum. The resulting block edges are shared across all of the burst's detectors. A dedicated fine-grid pass (1 ms cells with the Poisson block fitness, run on each burst's pulse core) shows that only 10% (11/106) of the sample contains significant sub-0.1 s structure—most prominently GRB 130427A, with credible blocks down to 2 ms—so intra-bin spectral evolution is a quantified, per-burst caveat for that fast minority rather than a sample-wide one. The provisional run yields 1057 time-resolved spectra, to be regenerated by the authoritative re-block on the approved backgrounds.

For each block, the spectral data are assembled in `3ML`: per detector, a source count spectrum is extracted over the block interval, the background count spectrum is the polynomial model (Section 2.2) integrated over the same interval with its Gaussian uncertainty, and the corresponding detector response matrix folds the model into count space. The detectors of a burst are then fit jointly over their respective energy ranges (Section 2.6); each block is one such joint fit.

2.4. *Block Definition and Grading*

We grade each burst by the quality of its time-resolved coverage, following a three-tier scheme:

1. *Gold*: 40 bursts with ≥ 10 well-constrained bins, suitable for per-burst parameter relations.
2. *Silver*: 44 bursts with 5–9 bins, contributing to population statistics with caveats.
3. *Bronze*: 22 bursts with ≤ 4 bins, too few for per-burst trends; included only in the pooled distributions.

The grade is carried as a column in Table 3 and is used in Section 5 to show that our main results hold on the cleanest (Gold) subsample.

2.5. *Spectral Models*

We fit each spectrum with six photon models spanning the two interpretations of the spectral curvature while remaining empirical and comparable with the GBM catalogs (e.g., Kaneko et al. 2006; Gruber et al. 2014). Empirical functions are flexible and model-agnostic; although they do not map uniquely onto a mechanism, the low-energy index α still discriminates between scenarios at high significance (e.g., Acuner et al. 2020).

The Band function (Band et al. 1993) is a smoothly-joined broken power law; below the break $E_b = (\alpha - \beta)E_\mathrm{p}/(2+\alpha)$,

$$N(E) = A\,(E/E_0)^\alpha \, \exp[-(2+\alpha)E/E_\mathrm{p}]\,, \quad (1)$$

joining $N(E) \propto E^\beta$ above E_b, with $E_0 = 100$ keV. The cutoff power law (CPL), $N(E) = A\,(E/E_0)^\Gamma \exp(-E/E_c)$ with $E_\mathrm{p} = (2+\Gamma)E_c$, is the $\beta \to -\infty$ limit. The smoothly broken power law (SBPL) joins two power-law segments with photon indices α and β at a break energy E_b, with the sharpness of the join an adjustable parameter; we use the GBM-catalog parameterization (Kaneko et al. 2006) as implemented in astromodels, with the break scale held fixed.

To represent the synchrotron interpretation we use the double smoothly broken power law (2SBPL; Ravasio et al. 2018, whose Equations 1–4 give the functional form): three power-law segments with photon indices $\alpha_1, \alpha_2, \beta$ joined at two break energies—a lower break x_b and an upper break x_p (the latter at the νF_ν peak)—each join having a sharpness parameter (n_1, n_2). We fit the astromodels implementation, whose free parameters are the normalization, the three indices, and

[1] The numbers in this paper use a provisional, automatically-selected background catalogue pending completion of this approval pass and the authoritative re-fit; the qualitative conclusions are not expected to be sensitive to the refinement.

the two break energies. Under the optically-thin synchrotron interpretation the two breaks are physical—x_b is the cooling frequency ν_c and $x_p = \nu_m$ the injection frequency—and the segment between them carries the synchrotron slopes $\alpha_1 \simeq -2/3$ (slow cooling) steepening toward $-3/2$ (fast cooling). These characteristic values follow from the radiation physics itself: the low-energy tail of even a single electron's synchrotron spectrum cannot be harder than $N(E) \propto E^{-2/3}$, and a fully cooled electron population softens it to $E^{-3/2}$, so measured indices above these "lines of death" challenge a pure synchrotron origin (Preece et al. 1998; Sari et al. 1998; Ravasio et al. 2019). We hold the break sharpnesses fixed ($n_1 = n_2 = 2$ in this provisional run; the verified re-fit will adopt the $n_1 = 5.38$, $n_2 = 2.69$ calibration of Ravasio et al. 2018, who found that leaving n_1 free does not converge reliably—the same behavior we observe). The full fit configuration (parameter bounds, seeds, and restart grids) is listed in Appendix A.

To represent the photospheric interpretation we add a blackbody to the two simplest non-thermal continua, following the multi-component decomposition of Guiriec et al. (2011, 2015), in which a sub-dominant quasi-thermal component accompanies the non-thermal emission and the two are fit *simultaneously*. The blackbody photon spectrum is $N(E) \propto E^2/[\exp(E/kT)-1]$, and we form the composites Band+BB and CPL+BB; we quantify the thermal contribution by the flux ratio $F_{\rm BB}/F_{\rm tot}$ (Guiriec et al. 2015). A BB+SBPL composite reproduces the curvature of a 2SBPL near the peak, so the +BB models act as a practical thermal proxy for the two-break model.

2.6. *Spectral Fitting and Model Selection*

All fits are performed in the Multi-Mission Maximum Likelihood framework (3ML; Vianello et al. 2015) with the `pgstat` likelihood, the appropriate statistic when the source counts are Poisson-distributed and the background level carries Gaussian uncertainties from the polynomial fit. Model comparison uses the Akaike and Bayesian information criteria (AIC, BIC; Akaike 1974; Schwarz 1978) computed from the standard $-2\ln L$. An information criterion is the fit's $-2\ln L$ penalized by the number of free parameters, so a lower value indicates a better description of the data after accounting for model complexity; a gap of $\Delta\mathrm{AIC} \geq 10$ between two models is conventionally decisive. Which statistic is *valid*, however, depends on the relationship between the models compared. For *nested* pairs—where the simpler model is a special case of the more complex one, as for Band+BB/Band, CPL+BB/CPL, and 2SBPL/SBPL—a blackbody or second break is deemed significant at $\Delta\mathrm{AIC} \geq 10$ relative to the parent continuum—the AIC analog of the $\Delta\mathrm{DIC} \geq 10$ threshold of Li et al. (2021) and the Bayes-factor threshold ($2\ln K > 10$) of Burgess et al. (2019), and equivalent here to a likelihood-ratio statistic (Wilks 1938) of $\geq$14, beyond the nominal 99% (2 d.o.f.) point; we prefer the information-criterion form because the χ^2 calibration of the classical LRT is only approximate when a component normalization sits at a parameter boundary. Every blackbody and second break that passes $\Delta\mathrm{AIC} \geq 10$ also passes the more conservative $\Delta\mathrm{BIC}$ criterion. For *non-nested* comparisons—in particular the central thermal (+BB) versus two-break (2SBPL) question, where neither model contains the other as a limiting case—the LRT is undefined, and we rely on $\Delta\mathrm{AIC}$ alone, with $\Delta\mathrm{BIC}$ as a cross-check. A physical-validity gate prevents a fit from *winning* the selection if any key parameter is railed against a bound or, for the 2SBPL, if the breaks are mis-ordered ($x_b \geq x_p$); in the 4% (43/1057) of bins where no model passes the gate we assign no preferred model. The blackbody possesses a railing local minimum, so we multi-start the +BB models whose blackbody is railed or insignificant from a grid of temperature seeds and retain the lowest $-2\ln L$. The 2SBPL has no analogous restart in the present (provisional) run, so two-break counts are conservative lower limits (Section 3.4). We quantify each parameter correlation by the Spearman rank coefficient ρ and a least-squares slope in log space. Energy and photon fluxes are obtained by integrating the best-fit Band model over 10–1000 keV, with uncertainties propagated by Monte Carlo over the fitted parameter uncertainties (200 draws; 16th–84th percentiles). For the principal ν_m–ν_c relation, whose two variables carry comparable uncertainties, we maximize the D'Agostini (2005) errors-in-both-variables likelihood,

$$-2\ln\mathcal{L} = \sum_i \left[\ln 2\pi\sigma_i^2 + \frac{(y_i - mx_i - c)^2}{\sigma_i^2}\right], \quad \sigma_i^2 = \sigma_{\rm sc}^2 + \sigma_{y,i}^2 + m^2\sigma_x^2 \tag{2}$$

with the measured break uncertainties (in dex) on both axes and the intrinsic scatter $\sigma_{\rm sc}$—the spread of the relation beyond what the measurement errors alone explain—a free parameter.

3. OBSERVATIONAL PROPERTIES

Throughout this section, each statistic is computed on the subset of the 1057 bins for which the relevant quantity is constrained: 792 bins (75%) have a physically valid Band fit (the reference for α, $E_{\rm p}$, and flux); 301 bins have a decisive blackbody ($\Delta\mathrm{AIC} \geq 10$; Section 2.6), of which 233 also have a valid Band flux; 349 bins have a valid 2SBPL with ordered breaks; 950 bins have at least two physically valid models (the model-comparison set); and per-burst statistics use the 100 bursts with at least one valid Band bin, with per-burst correlations requiring ≥5 bins (68 bursts) or ≥4 $E_{\rm p}$–kT pairs (19 bursts).

3.1. *Parameter Distributions*

We now examine the distributions of the spectral parameters (Figure 1, Table 4). The low-energy index has a median $\alpha = -0.71 \pm 0.49$, the peak energy $\log_{10}(E_{\rm p}/{\rm keV}) = 2.28 \pm 0.40$ (median $E_{\rm p} \simeq 192$ keV), and the high-energy index a median $\beta = -2.6$, in agreement with previous catalogs (e.g., Kaneko et al. 2006; Gruber et al. 2014). The blackbody temperature, where decisively required, has a median $kT \simeq 28$ keV and spans $kT \approx 3$–70 keV (5th–95th percentiles), bracketing the ∼40 keV temperatures of Guiriec et al. (2015). K-S tests between tiers (Table 5) show that the Gold-tier distributions differ modestly from the lower tiers ($p \approx 0.004$ for α, 0.03 for $\log E_{\rm p}$), as expected since the brightest bursts populate the Gold tier; the difference does not affect the conclusions below.

3.2. *Spectral Evolution*

We classify the $E_{\rm p}$ evolution into hard-to-soft (HTS) and intensity-tracking (IT) patterns following the physical discriminant of Lu et al. (2012) and Basak & Rao (2014): the two classes differ in the *rising* phase of the pulse, where an IT pulse hardens with the flux while an HTS pulse is already at its hardest and softening (in the decay phase both soften together, so a whole-pulse $E_{\rm p}$–flux correlation cannot separate them). We therefore classify on the pre-peak behavior—the sign of the $E_{\rm p}$ trend through the flux peak, falling back to the position of the $E_{\rm p}$ maximum when the rise is covered by fewer than three bins. Of the 76 bursts with ≥4 classifiable bins, 63% (48/76) are HTS, 25% (19/76) are IT, and 12% (9/76) are ambiguous (Figure 2), an HTS-dominated mix consistent with the small single-pulse samples of Lu et al. (2012) and Basak & Rao (2014). The general trend is that pulses soften over time, with α decreasing through the pulse.

3.3. *Parameter Correlations*

Turning to the parameter correlations, we fit each pair with the functional forms adopted by Li et al. (2021) and report both the per-burst incidence and the pooled, sample-wide trend (Table 6).

3.3.1. *The F–α Relation*

We fit $F = F_0\, e^{k\alpha}$ and find a positive correlation in 84% (57/68) of bursts, with pooled $k = 0.27$: brighter spectra tend to have harder low-energy indices, though the trend is weak (pooled $\rho = 0.17$).

3.3.2. *The F–$E_{\rm p}$ Relation*

We fit a power law $F \propto E_{\rm p}^{k_1}$ and find $k_1 = 0.66$, positive in 94% (64/68) of bursts (pooled $\rho = 0.43$). This correlation, however, is partly *induced*: the energy flux is $\int E\, N(E)\, dE$, which rises when $E_{\rm p}$ shifts to higher energies even at a fixed photon rate, so part of any F–$E_{\rm p}$ correlation is built into the quantities themselves. In photon flux (which carries no such energy weighting) the rank correlation collapses to $\rho = 0.15$, consistent with the absence of an intrinsic $E_{\rm p}$–luminosity relation (Mei et al. 2025).

3.3.3. *The α–$E_{\rm p}$ Relation*

We fit $\alpha = k_2 \log_{10} E_{\rm p} + c$ and find only a weak positive trend, $k_2 = 0.02$ (pooled $\rho = 0.10$); a positive per-burst correlation appears in 66% (45/68) of bursts but is individually significant in only a handful.

3.4. *Curvature and the Two Pictures*

We now turn to the curvature beyond a single spectral break, the central question of this work. Restricting to bins requiring such curvature ($\Delta{\rm AIC} > 6$ relative to the best single-break model, with the curved model physically valid), we classify each as thermal (+BB preferred) or two-break (2SBPL preferred with $\Delta{\rm AIC} \geq 10$ over the SBPL); the remainder are degenerate between the two pictures (Table 1). Of the 166 such bins, 91% (151/166) are thermal-or-degenerate and 9% (15/166) decisively prefer the explicit second break (87%/13% at the stricter $\Delta{\rm AIC} \geq 10$ curvature cut); because the 2SBPL carries no convergence restart in this run (Section 2.6), the two-break fractions are lower limits. More broadly, no single empirical model is decisively preferred over the next-best alternative ($\Delta{\rm AIC} < 10$) in 92% (875/950) of the bins with at least two valid models—and all valid models agree within $\Delta{\rm AIC} < 10$ in 58%—echoing the difficulty Basak & Rao (2014) noted in separating thermal and non-thermal descriptions on goodness-of-fit alone. Where a blackbody is decisively required, its flux fraction is typically small but spans a wide range ($F_{\rm BB}/F_{\rm tot}$ median 0.006, interquartile 0.001–0.10): the brightest bins permit detection of percent-level blackbodies, so the median reflects detectability as much as physics, with the upper quartile matching the 1–10% energy fractions of Guiriec et al. (2015). Testing the Burgess et al. (2014a) relation per burst on the decisive-blackbody sample, the $E_{\rm p}$–kT correlation is positive in 89% (17/19) of testable bursts (12 individually significant), most cleanly in the highest signal-to-noise bursts (GRB 130427A: $\rho = 0.93$,

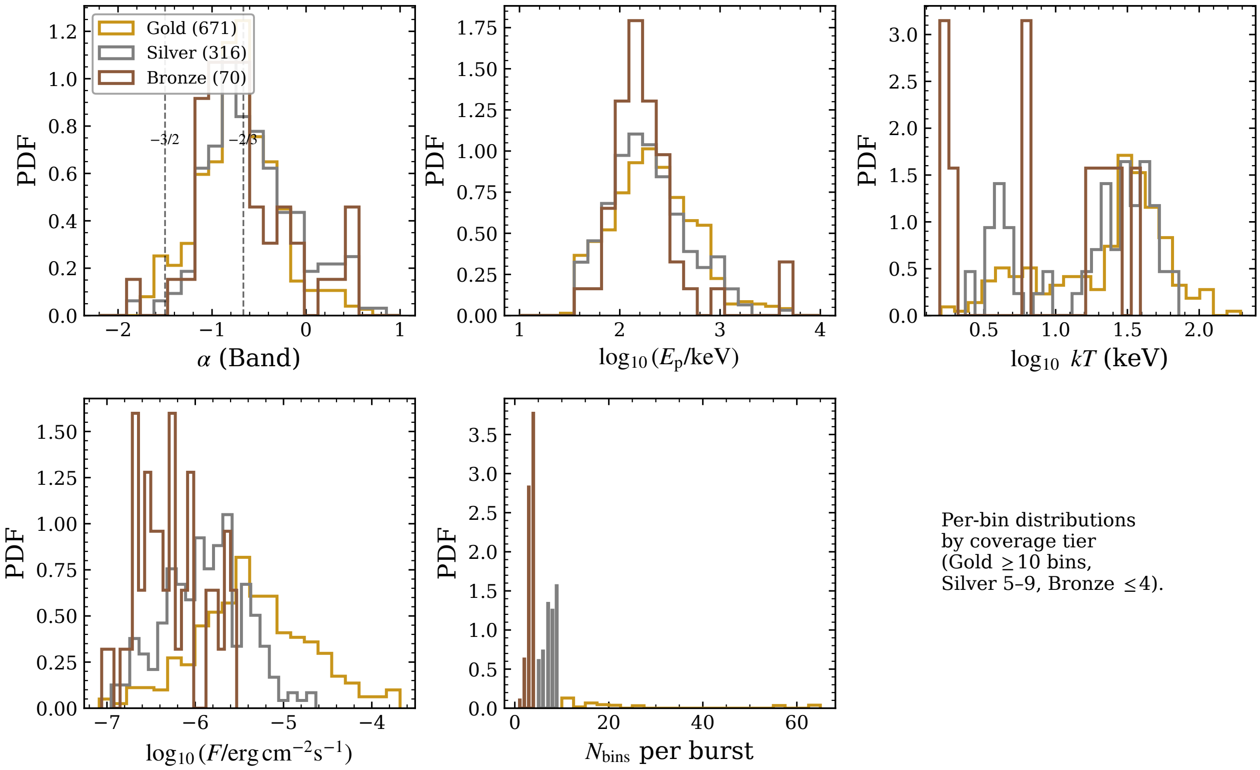

Figure 1. Distributions of the Band low-energy index α, peak energy $E_{\rm p}$, blackbody temperature kT, energy flux F, and the number of bins per burst, split by coverage tier (Gold/Silver/Bronze). The legend gives the total number of time bins per tier (1057 in all); each panel shows the subset in which the plotted parameter is constrained. Dashed lines in the α panel mark the slow- and fast-cooled synchrotron limits ($-2/3$, $-3/2$).

$E_{\rm p} \propto kT^{0.88\pm0.07}$; GRB 160625B: $\rho = 0.78$); fainter bursts yield too few decisive-blackbody bins to test, a consequence of the empirical continuum absorbing the thermal curvature. Finally, the two 2SBPL breaks are positively correlated ($\rho = 0.57$, $N = 349$): the D'Agostini (2005) fit gives $\nu_m \propto \nu_c^{0.46\pm0.05}$ with intrinsic scatter $\sigma_{\rm sc}$= 0.38 dex, and, unlike a pure inter-burst artifact, the relation is significant both within ($\rho = 0.43$) and between ($\rho = 0.51$) bursts (Section 5.2).

4. ASSESSMENT OF COMPATIBILITY WITH EMISSION MODELS

4.1. *Spectral Shape*

Optically-thin synchrotron limits the low-energy index to $\alpha \leq -2/3$ (slow cooling), steepening to $-3/2$ (fast cooling), whereas a photosphere can be harder; a photospheric preference can be claimed with confidence only for $\alpha > -0.5$ at high significance (Acuner et al. 2020). We place the maximum α of each burst against $E_{\rm p}$ and the synchrotron and photospheric limiting lines (Figure 5). We find that 74% (74/100) of bursts reach a peak α above the slow-cooling synchrotron line

Table 1. Classification of the bins requiring curvature beyond a single spectral break.

	ΔAIC > 6	ΔAIC ≥ 10
Curvature-required bins	166	108
thermal-or-degenerate	151 (91%)	94 (87%)
two-break (2SBPL, decisive)	15 (9%)	14 (13%)

NOTE—Curvature is required when the best valid curved model (+BB or 2SBPL) improves on the best valid single-break model by the stated ΔAIC; "decisive" two-break additionally requires ΔAIC ≥ 10 for the 2SBPL over the SBPL. Two-break fractions are lower limits (Section 2.6).

($-2/3$) and 64% (64/100) enter the photospheric region ($\alpha > -0.5$), broadly consistent with the hard low-energy indices reported by Basak & Rao (2014) and Crider et al. (1997) and difficult to reconcile with pure fast-cooled synchrotron.

4.2. *Spectral Evolution and Correlations*

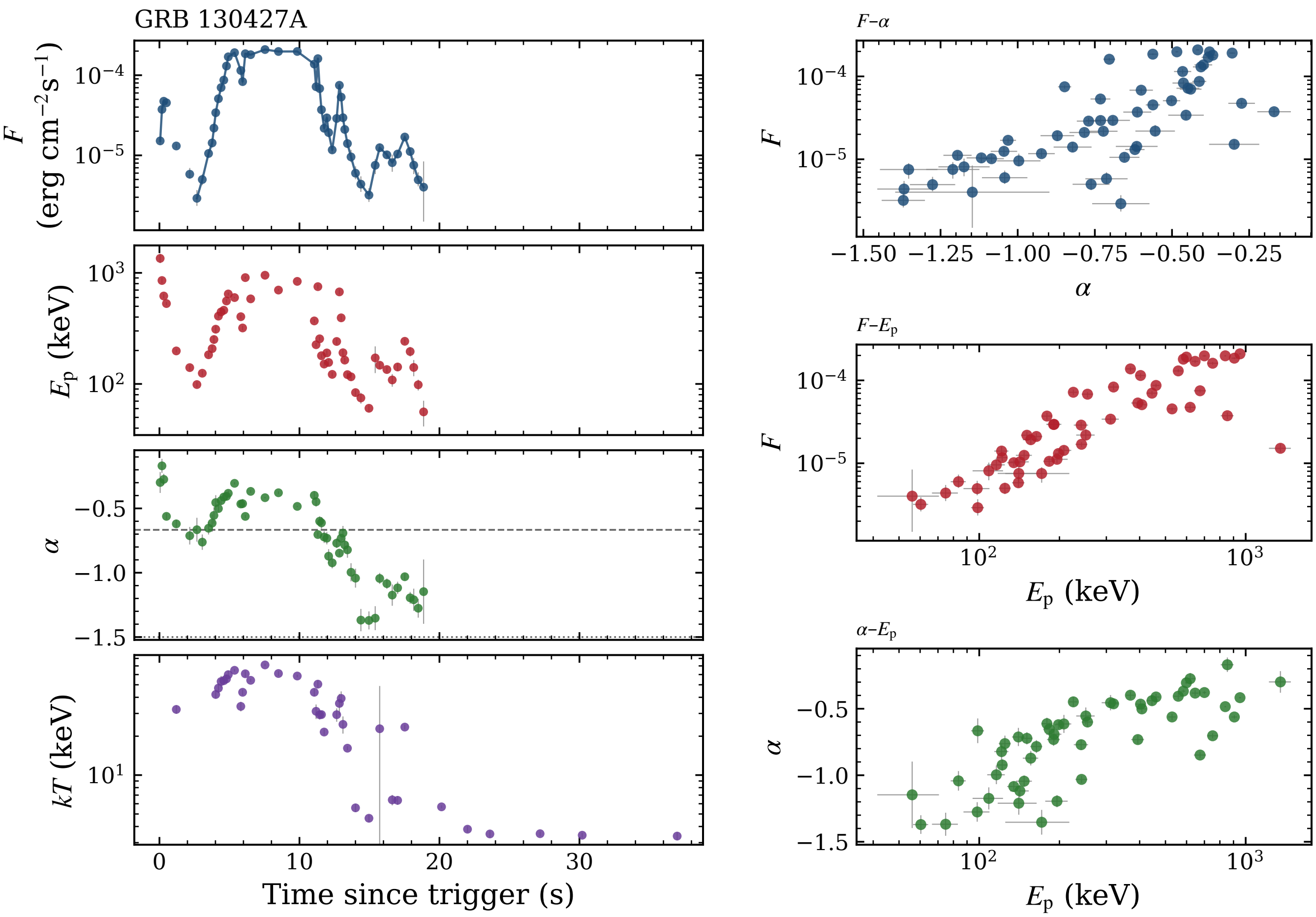

Figure 2. Time-resolved spectral evolution of GRB 130427A (65 bins in all). Left, top to bottom: the energy-flux light curve $F(t)$, $E_\mathrm{p}(t)$, $\alpha(t)$ (dashed/dotted lines at $-2/3$, $-3/2$), and $kT(t)$. Each panel shows the bins in which the plotted quantity is constrained (54 Band-valid bins for F, E_p, α; decisive-blackbody bins for kT, which extend into the late tail). Right: the F–α, F–E_p, and α–E_p relations for the same burst.

The curvature split (Section 3.4) indicates that most single-pulse curvature is statistically consistent with an added blackbody, with a minority decisively requiring a second synchrotron break. The sub-dominant $F_\mathrm{BB}/F_\mathrm{tot}$, the recovery of the E_p–kT correlation only at high signal-to-noise, and the survival of the kT–intensity correlation in photon flux ($\rho = 0.24$, against 0.49 in energy flux, where the E_p–intensity correlation collapses instead) together indicate that the empirical continuum can mask a sub-dominant thermal component (cf. Basak & Rao 2014; Burgess et al. 2014b).

5. DISCUSSION

5.1. *Comparison with Previous Works on Single-Pulse GRBs*

Single-pulse bursts have long been the preferred laboratory for prompt-emission spectroscopy, and our sample of 106 bursts is, to our knowledge, the largest uniformly-analyzed set to date. Table 7 summarizes the comparison; we discuss it thematically.

The single-pulse lineage and spectral evolution. Ryde (2004) and Ryde & Pe'er (2009) established that a single FRED-like pulse exhibits the simplest spectral evolution and is well described by a blackbody-plus-power-law, making single pulses the cleanest test of the prompt mechanism. Lu et al. (2012) classified the E_p evolution of bright single-pulse Fermi bursts into hard-to-soft (HTS) and intensity-tracking (IT) patterns, a scheme extended by Basak & Rao (2013, 2014). Our rise-phase classification (HTS 63%, IT 25%; Section 3.2) recovers the same two modes and the same HTS-dominated balance as those small samples. The agreement on the low-energy index is also notable: our median Band $\alpha = -0.71$ matches the IT mean (-0.68) of Basak & Rao (2014) and is close to the slow-cooled value -0.81 ± 0.1 of Burgess et al. (2014b). On the synchrotron line of death the comparison must be made carefully: Basak

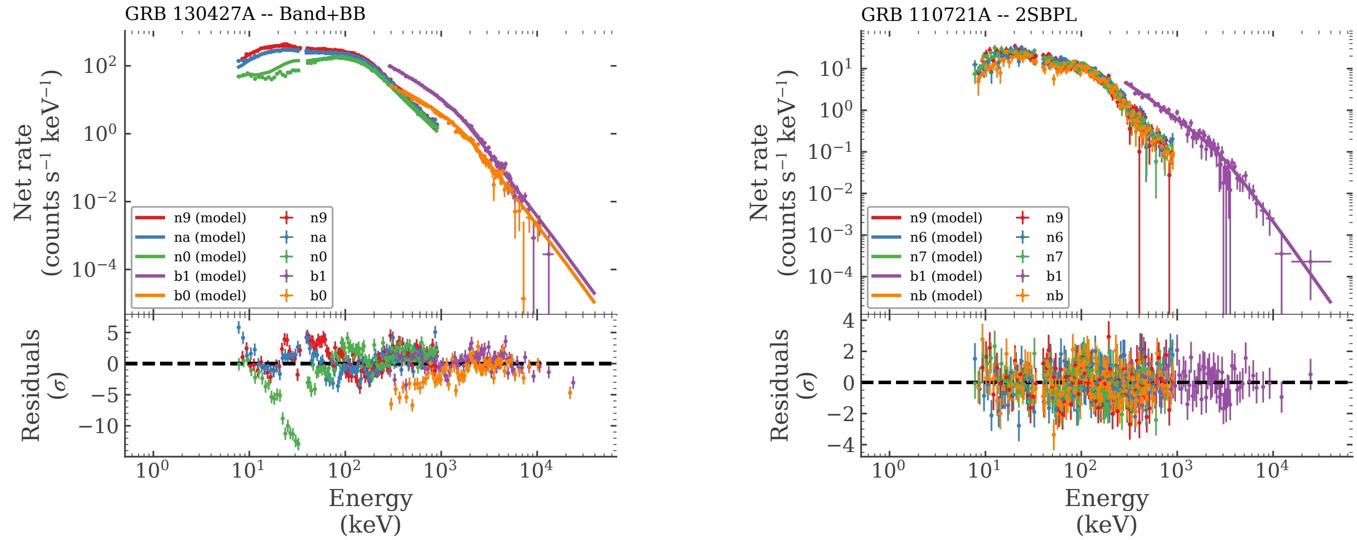

Figure 3. Count spectra and residuals for the two emblematic curvature cases. Left: a decisively blackbody-requiring bin of GRB 130427A (8.7–11.0 s) fit with Band+BB, chosen away from the flux peak where pile-up may affect this exceptionally bright burst (cf. Mei et al. 2025); the narrow low-energy residual structure in one NaI is of the kind attributed to response calibration and intra-bin spectral evolution in very bright bursts (cf. Ronchi et al. 2020). Right: the most decisive two-break bin of GRB 110721A (0.5–2.0 s) fit with the 2SBPL, with flat residuals across 8 keV–100 MeV (NaI, BGO, and LLE). Each color is one detector; solid lines show the folded model.

& Rao (2014) report $\alpha > -2/3$ in 63.4% of their HTS *bins* (74.6% excluding GRB 110721A), whereas our like-for-like per-bin fraction is 44% over a sample dominated by IT-era decay bins, and 74% (74/100) of our *bursts* exceed the line in at least one bin. The numbers are therefore comparable rather than identical, but all the single-pulse studies—including Basak & Rao (2013) and Guiriec et al. (2010)—find the line of death frequently violated, which we now establish at population scale.

The thermal / multi-component program. A sub-dominant photosphere has been advocated through blackbody composites (Ryde 2004; Guiriec et al. 2010, 2011, 2015; Basak & Rao 2014). Basak & Rao (2014) find that a blackbody-plus-power-law peaks at $\sim 3kT$, below the Band νF_ν peak, and favor multi-blackbody models over Band; Guiriec et al. (2015) require a cutoff-power-law+blackbody+power-law for two bright GRBs, with the blackbody carrying only 1–10% of the energy ($kT \sim 40$ keV) and argue that the hard α values are an artifact of fitting Band to a multi-component spectrum. Our results support this picture at scale: the thermal flux fraction is sub-dominant (median 0.006 with interquartile range 0.001–0.10, the upper quartile matching their 1–10% energy range), our decisive blackbody temperatures ($kT \approx 3$–70 keV, median 28) bracket their ∼40 keV, and 91% (151/166) of curvature-requiring bins are equally or better described by adding a sub-dominant blackbody than by a second synchrotron break.

The synchrotron / physical-model program. Burgess et al. (2014a) fit a physical synchrotron+blackbody model to six single-pulse bursts and reported a strong thermal–non-thermal correlation ($E_\mathrm{p} \propto kT^\alpha$, pooled $\rho \approx 0.8$, $\alpha \sim 1$–2); we test this at ∼18× the sample size and recover it in 89% (17/19) of the bursts with enough decisive-blackbody bins to test, most cleanly in the highest-S/N bursts (GRB 130427A: $\rho = 0.93$, slope 0.88 ± 0.07), consistent with empirical continua absorbing the thermal curvature in fainter bursts. The double-break synchrotron picture (Oganesyan et al. 2017; Ravasio et al. 2018, 2019) predicts the 2SBPL we fit; in the brightest GRB ever, GRB 221009A, Ravasio et al. (2023) find a genuine second low-energy break only in the highest-S/N intervals—directly mirroring our result that an explicit second break wins decisively in just 9% (15/166) of curvature bins (a lower limit), epitomized by GRB 110721A. This is expected: Ravasio et al. (2018) showed that the +BB and two-break descriptions separate only in bright spectra with a large peak-to-break ratio ($E_\mathrm{peak}/E_\mathrm{break} \gtrsim 10$), precisely the high-S/N regime. Finally, Mei et al. (2025) find that the synchrotron model is acceptable for most bursts, that they lie in a fast/intermediate-cooling regime ($\nu_m \gtrsim \nu_c$), and that the spectral–energetics link is ν_c–L_iso rather than the Yonetoku relation. We independently support the absence of an E_p–luminosity relation: the apparent E_p–intensity correlation (energy-flux $\rho = 0.43$) collapses in photon flux ($\rho = 0.15$), while the kT–intensity corre-

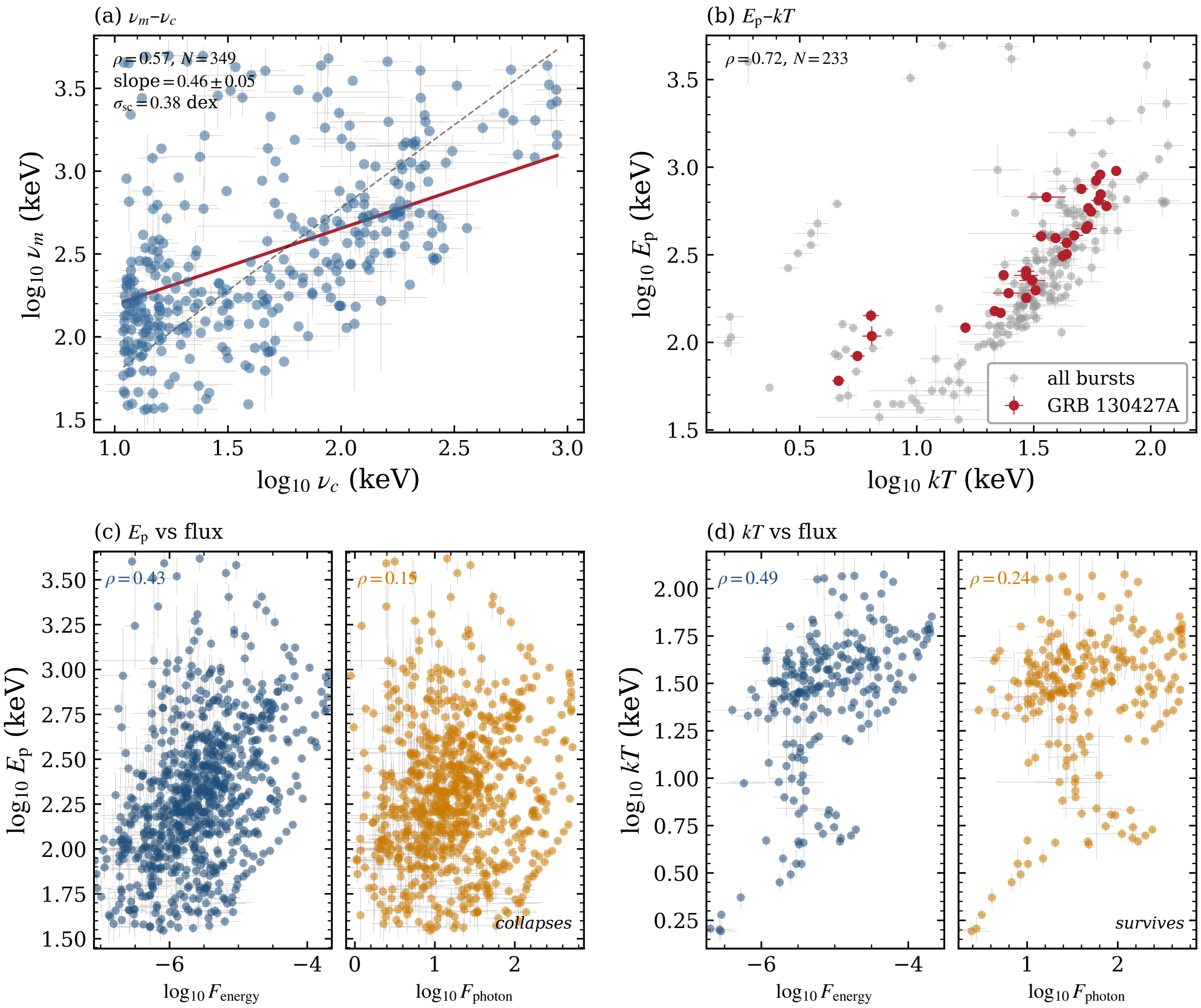

Figure 4. Spectral-break and temperature correlations. (a) The two 2SBPL break energies ν_m vs ν_c ($N = 349$; red: D'Agostini 2005 errors-in-both-variables fit, slope 0.46 ± 0.05; dashed: 1:1). (b) The E_p–kT relation over the decisive-blackbody bins, with GRB 130427A highlighted. (c) E_p vs energy flux (left, blue) and photon flux (right, orange), sharing the y-axis: the correlation *collapses* from $\rho = 0.43$ to 0.15, showing it is largely induced by energy weighting. (d) The same comparison for kT: the correlation *survives* in photon flux ($\rho = 0.24$; energy flux 0.49), indicating a real thermal–intensity link. Error bars are drawn where the fitted parameter is constrained (relative uncertainty below 100%; flux interval below 0.5 dex).

lation survives ($\rho = 0.24$); and our 2SBPL bins, with $\nu_c < \nu_m$ by construction, sit in the same marginally-fast-cooling regime.

Large catalogs and the degeneracy lesson. Compared with the broad GBM catalogs (Kaneko et al. 2006; Gruber et al. 2014; Yu et al. 2016) and the short-GRB Bayesian catalog of Burgess et al. (2019)—in which 513/525 short-burst spectra prefer a cutoff power law with no thermal component—our six-model analysis adds an explicit thermal-versus-two-break decomposition. Our central caution echoes Basak & Rao (2014): no single empirical model is decisively preferred over the next-best alternative in 92% (875/950) of comparable bins, so distinguishing the emission mechanism requires physical models (Burgess et al. 2014b) or higher signal-to-noise, not empirical goodness-of-fit alone. Throughout, we verify our main results on the cleanest (Gold) subsample.

5.2. *The Two-Break Relation*

The pooled ν_m–ν_c correlation ($\rho = 0.57$, $N = 349$) can be decomposed into two distinct statements: a *between*-burst correlation (bursts whose ν_c is higher also have higher ν_m on average) and a *within*-burst correlation (the two breaks move together in time inside an individual burst). Both hold. Between bursts, $\rho = 0.51$; within bursts, beyond the pooled mean-subtracted residuals ($\rho = 0.43$), the cluster-honest per-burst test gives

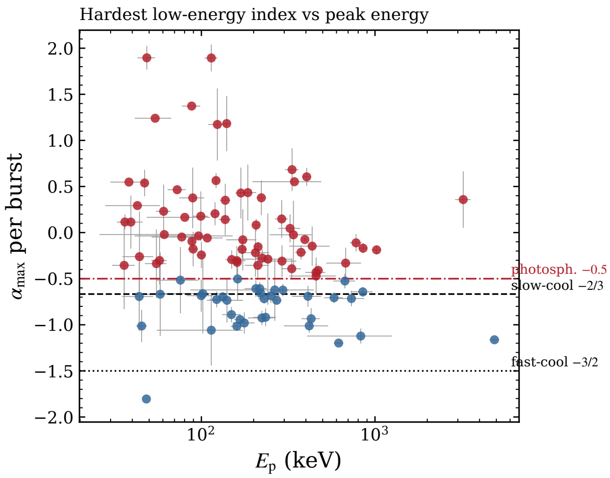

Figure 5. Hardest low-energy index per burst, $\alpha_{\max}$, vs $E_{\rm p}$, with the slow- and fast-cooled synchrotron lines ($-2/3$, $-3/2$) and the photospheric limit (-0.5). Red points (64/100) reach the photospheric region in at least one bin. The few extreme values ($\alpha_{\max} \gtrsim 1$) arise from weakly constrained bins and carry correspondingly large uncertainties.

a median per-burst $\rho = 0.50$, positive in 41/54 bursts with $\geq$3 double-break bins (sign-test $p = 2 \times 10^{-4}$). The relation is therefore not merely an inter-burst trend: harder bursts have both breaks at higher energies, and within a burst the two breaks move together. Its slope, however, depends on the estimator because the intrinsic scatter is large: the D'Agostini (2005) fit gives 0.46 ± 0.05 ($\sigma_{\rm sc}$= 0.38 dex), restricting to bins with both breaks constrained gives 0.64 ± 0.05 ($\rho = 0.67$, $N = 253$), and symmetric unweighted regression gives 1.05 ± 0.06; we therefore regard the *existence* of the correlation, not its slope, as the robust result. It is also robust to the fit boundary: excluding the 26% of ν_c values within a factor of two of the 8 keV NaI edge leaves $\rho = 0.55$ ($N = 260$). Because ν_c and ν_m are breaks of the same spectrum fit simultaneously, part of the within-burst correlation can be induced by their joint estimation; we note, however, that time-resolved *physical* synchrotron fits of GRB 180720B show the cooling and injection energies decreasing together through the burst with $E_{\rm m}/E_{\rm c} \approx$ 13–22 (Ronchi et al. 2020)—the same co-evolution, recovered with a model in which the two breaks are not independently adjustable shape parameters. The physically meaningful quantity is the ratio ν_m/ν_c (the cooling regime). Since the 2SBPL fit enforces $\nu_c < \nu_m$, the bins with a measurable double break sit in the fast-cooling regime by construction; quantitatively, the ratio distribution has a median $\nu_m/\nu_c \simeq 6$ (interquartile 3–13), with 21% (75/349) in the intermediate regime $1 < \nu_m/\nu_c < 3$ of Mei et al. (2025)—close to the ~25% they find in their time-resolved sample, and well below the ~60% of their single-peak-bin sample. Mei et al. (2025) use this ratio as a cooling-regime classifier rather than fitting a break–break relation; within the single burst GRB 160625B, Ravasio et al. (2018) found $E_{\rm peak}$–$E_{\rm break}$ positively correlated ($\rho = 0.61$), consistent in sign with our within-burst result. Our positive ν_m–ν_c correlation is, to our knowledge, the first such relation reported for a large single-pulse *sample*, and warrants confirmation with physical synchrotron fits.

5.3. *Implication for the Radiation Process*

The low-energy index peaking near $\alpha \approx -0.7$, between the synchrotron limits but with a substantial fraction of bursts above them, is consistent with marginally-fast/slow-cooled synchrotron with ongoing acceleration in some bursts and a photospheric contribution in others (cf. Basak & Rao 2014; Burgess et al. 2014b). The $E_{\rm p}$–kT slope can be mapped to the outflow dynamics because the bulk Lorentz factor of an accelerating jet grows with radius as $\Gamma \propto r^{\mu}$, with $\mu = 1$ for a baryon-dominated fireball and $\mu = 1/3$ for a magnetically-accelerated jet; the acceleration law sets how the photospheric temperature declines relative to the non-thermal peak, so $\alpha \sim 1$ indicates a baryon-dominated and $\alpha \sim 2$ a magnetically-dominated jet (Burgess et al. 2014a). The slope we recover for the cleanest case, GRB 130427A ($E_{\rm p} \propto kT^{0.88 \pm 0.07}$, D'Agostini 2005 fit), is consistent with the baryon-dominated value, but we stress that with empirical (not physical) models this mapping is only indicative. The clearest two-break burst in our sample, GRB 110721A (2SBPL-preferred, $F_{\rm BB}/F_{\rm tot} \approx 0.001$), is the counter-example in which the curvature is genuinely a second synchrotron break rather than a thermal component.

6. SUMMARY

In this paper, we performed time-resolved empirical spectroscopy of 106 single-pulse, bright Fermi/GBM bursts (1057 spectra; six models each), one of the largest uniform single-pulse spectral surveys to date. Our main findings are as follows.

1. The low-energy index peaks at $\alpha \approx -0.7$, near the slow-cooled synchrotron limit, with 74% (74/100) of bursts reaching harder values in at least one bin; the rise-phase $E_{\rm p}$ evolution is HTS-dominated (63%, 48/76), as in previous single-pulse samples.

2. Of the bins requiring curvature beyond a single break, 91% (151/166) are equally or better described by adding a blackbody than by an explicit second break (the 9% two-break fraction is a

Table 2. Global properties of the sample (representative rows).

GRB	T_{90} (s)	Fluence	NaI	LAT	$N_{\rm bins}$
130427A	138.2	2.5×10^{-3}	n6	Y	65
160625B	453.4	6.4×10^{-4}	n6	Y	57
150902A	13.6	8.3×10^{-5}	n0	Y	19
110721A	21.8	3.7×10^{-5}	n6	Y	11
081125A	9.3	1.9×10^{-5}	na	N	8

NOTE—Fluence in erg cm^{-2} (10–1000 keV, GBM catalog). The full machine-readable table lists all 106 bursts; selection is on pulse shape, not duration.

lower limit); no single model is decisively preferred over the next-best in 92% (875/950) of comparable bins.

3. The Burgess et al. (2014a) $E_{\rm p}$–kT correlation is recovered in 89% (17/19) of the bursts with enough decisive-blackbody bins to test (GRB 130427A: $\rho = 0.93$, $E_{\rm p} \propto kT^{0.88\pm0.07}$), but fainter bursts cannot be tested with empirical models—a model-sensitivity, not a sample, limit.

4. The two 2SBPL break energies are positively correlated ($\rho = 0.57$; D'Agostini 2005 slope 0.46 ± 0.05, $\sigma_{\rm sc}= 0.38$ dex), both within and between bursts.

We conclude that the dominant spectral curvature of bright single-pulse GRBs is more economically described by a sub-dominant photospheric component than by a second synchrotron break, with a genuine two-break minority—epitomized by GRB 110721A—in which the cooling and injection frequencies are correlated.

This work made use of 3ML (Vianello et al. 2015) and the Fermi/GBM and LAT public data; Bayesian blocks follow Scargle et al. (2013) and correlation fits the D'Agostini (2005) method.

Facilities: Fermi(GBM), Fermi(LAT)

Software: 3ML (Vianello et al. 2015), Astropy, Matplotlib, NumPy, SciPy, emcee (Foreman-Mackey et al. 2013)

Table 3. Block grading and preferred model (representative rows).

GRB	N bins	Grade	Preferred model	$F_{\rm BB}/F_{\rm tot}$
130427A	65	Gold	Band+BB	0.12
160625B	57	Gold	Band+BB	0.004
150902A	19	Gold	Band+BB	0.13
110721A	11	Gold	2SBPL	0.001
081125A	8	Silver	Band+BB[a]	0.002

NOTE—Preferred model is the most frequent physically valid AIC winner across a burst's bins ([a]tied with Band); $F_{\rm BB}/F_{\rm tot}$ is the per-burst median over decisive-blackbody (ΔAIC $\geq$ 10) bins. The full per-burst table and the complete per-bin fit catalog (1057 rows: times, parameters with uncertainties, information criteria, and likelihood-ratio statistics for all six models) are available in machine-readable form.

Table 4. Band-parameter means ± standard deviation per tier.

Tier	$N_{\rm GRB}$	$N_{\rm spec}$	$\langle\alpha\rangle$	$\langle\log_{10} E_{\rm p}\rangle$
Gold	40	521	-0.71 ± 0.43	2.34 ± 0.41
Silver	44	226	-0.52 ± 0.58	2.28 ± 0.38
Bronze	22	45	-0.66 ± 0.51	2.27 ± 0.38

NOTE—Pooled blackbody temperature over the 301 decisive-blackbody bins (ΔAIC $\geq$ 10): median kT = 28.5 keV (5–95%: 3.4–70 keV).

Table 5. K-S comparison of tier distributions (Gold vs. Silver+Bronze).

	α		$\log_{10} E_{\rm p}$	
	D	p	D	p
Gold vs. rest	0.13	0.004	0.11	0.031

APPENDIX

Table 6. Pooled correlation results across all valid time bins.

Relation (y vs x)	N	ρ	p	slope	$\sigma_{\rm sc}$ (dex)
F vs α	792	0.17	3×10^{-6}	$k = 0.27$	0.65
F vs $E_{\rm p}$	792	0.43	$< 10^{-36}$	0.66	0.59
α vs $E_{\rm p}$	792	0.10	3×10^{-3}	0.02	$0.49^{\dagger}$
$E_{\rm p}$ vs kT	233	0.72	3×10^{-38}	0.56	0.36
ν_m vs ν_c	349	0.57	$< 10^{-30}$	0.46 ± 0.05	0.38
kT vs flux	233	0.24	2×10^{-4}	0.18	0.36

NOTE—ρ is the Spearman coefficient. Slopes are least-squares log–log fits (k is the e-folding index of $F = F_0 e^{k\alpha}$); $\sigma_{\rm sc}$ is the residual rms in dex, except † which is in units of α. The ν_m–ν_c slope and $\sigma_{\rm sc}$ are the D'Agostini (2005) errors-in-both-variables values with free intrinsic scatter (unweighted orthogonal regression gives 1.05 ± 0.06; least-squares 0.56). kT rows use the decisive-blackbody bins with valid Band flux. Energy fluxes are observer-frame (no redshifts). F–$E_{\rm p}$ uses energy flux and is partly induced (photon-flux $\rho = 0.15$); kT–flux uses photon flux (energy-flux $\rho = 0.49$). The ν_m–ν_c relation holds within bursts ($\rho = 0.43$ pooled; median per-burst $\rho = 0.50$) and between bursts ($\rho = 0.51$).

Table 7. Comparison with previous single-pulse / prompt-emission spectral studies.

Study	Sample	Models	Key finding and relation to this work
Ryde & Pe'er (2009)	BATSE single pulses	BBPL	Photospheric BBPL; single pulses as the cleanest laboratory (rationale we adopt).
Lu et al. (2012)	51+11 GBM	Band	HTS/IT $E_{\rm p}$-evolution classes; we recover both, HTS-dominated (HTS 63%/IT 25%).
Basak & Rao (2013)	19 GRB/43 pulses	Band	Pulse-wise Amati; α sometimes $> -2/3$.
Basak & Rao (2014)	9 single-pulse	BBPL/mBBPL/2BBPL	BB peaks $\sim 3kT$ below the Band peak; $\alpha > -2/3$ in 63.4% of HTS bins; favor multi-BB. Comparable to our per-bin 44% / per-burst 74%, at $\sim$12× the scale.
Burgess et al. (2014a)	6 single-pulse	synch.+BB	$E_{\rm p}$–kT correlation ($\rho \approx 0.8$); we recover it in 17/19 testable bursts (130427A $\rho = 0.93$).
Burgess et al. (2014b)	8 single-pulse	physical synch.	Slow-cooled $\alpha = -0.81$; BB in 5/8; fast cooling excluded.
Guiriec et al. (2015)	2 bright	C+BB+PL	Sub-dominant BB (1–10% energy, $kT \sim 40$ keV); hard α a Band-fit artifact. Our decisive-BB bins: $F_{\rm BB}/F_{\rm tot}$ up to $\sim$0.1, $kT \approx 3$–70 keV.
Ravasio et al. (2023)	GRB 221009A	SBPL/2SBPL	Genuine 2nd break only at highest S/N; mirrors our 9% two-break lower limit (e.g. 110721A).
Mei et al. (2025)	74 single-bin + 13 t.r.	physical synch.	Fast/intermediate cooling; ν_c–$L_{\rm iso}$, no Yonetoku. We confirm $E_{\rm p}$–flux is induced.
Burgess et al. (2019)	321 short	Band/CPL	Short bursts CPL-dominated (513/525); no thermal component.
This work	**106 single-pulse**	**6 empirical**	91% **thermal-or-degenerate vs** 9% **2nd break (lower limit);** 92% **(875/950) without a decisive winner;** ν_m–ν_c **correlated.**

Notes. "t.r." = time-resolved; all samples are Fermi/GBM (±LAT) unless noted. This work extends the single-pulse approach to the largest uniform sample.

Table 8. Free parameters, allowed ranges, and seeds.

Model	Parameter	Range	Seed
Band	α	$(-1.9, 1.9)$	-1.0
	$E_{\rm p}$ (keV)	(30, 5000)	200
	β	$(-5, -1.6)$	-2.25
CPL	Γ	$(-2, 1)$	-1.0
	E_c (keV)	$(10, 5 \times 10^4)$	200
SBPL	α	$(-2.5, 1.5)$	-1.0
	$E_{\rm b}$ (keV)	(10, 5000)	300
	β	$(-5, -1.5)$	-2.25
2SBPL	α_1	$(-2.5, 2.5)$	-0.66
	x_b (keV)	(10, 900)	50
	α_2	$(-3, 0.5)$	-1.5
	x_p (keV)	(30, 5000)	200
	β	$(-5, -1.5)$	-2.25
BB	kT (keV)	(1, 200)	30

Note—Normalizations are free with broad ranges. SBPL break scale and 2SBPL sharpnesses ($n_1 = n_2 = 2$) are held fixed in this run (Section 2.5). The x_b upper bound reflects the NaI band.

A. FIT CONFIGURATION

All fits use MINUIT within 3ML, each time bin seeded from the burst's time-integrated fit. Table 8 lists the free parameters, allowed ranges, and default seeds for each model component. Cross-detector effective-area constants are free within 0.8–1.2 (frozen to unity for the reference NaI). The blackbody multi-start re-fits each +BB model whose blackbody is railed or insignificant from three seeds (component defaults with $kT = 30$ keV; the time-integrated seed with $kT = 30$ and with 80 keV), retaining the lowest $-2\ln L$. The physical-validity gate declares a fit non-physical when any key shape parameter lies within 0.1% of its allowed span from a bound, or when the 2SBPL breaks are mis-ordered ($x_b \geq x_p$); non-physical fits remain in the catalog but cannot be selected as the preferred model, and in 4% (43/1057) of bins no model passes the gate and no preferred model is assigned. The complete per-bin fit catalog (1057 rows) and the full per-burst sample table are available as machine-readable tables.

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